

# Homework 1

## *Anharmonic Oscillator*

1. Let us use this problem to remind us of Euler's equation (which you should never forget):

$$e^{ix} = \cos(x) + i \sin(x)$$

In class we recalled that the general solution to the simple harmonic oscillator is:

$$x(t) = x_o \cos \omega t + \frac{v_o}{\omega} \sin \omega t = \alpha_1 e^{i\omega t} + \alpha_2 e^{-i\omega t}$$

where  $x_o = x(0)$  is the initial position and  $v_o = \dot{x}(0)$  is the initial velocity.

- Solve for  $\alpha_1$  and  $\alpha_2$  in terms of  $x_o, v_o$ .
- Prove that one can write the general solution as  $x(t) = A \cos(\omega t + \delta)$ , where  $A$  and  $\delta$  are real constants.
- Prove the following:

$$A = \sqrt{x_o^2 + \frac{v_o^2}{\omega^2}} \qquad \delta = -\arctan\left(\frac{v_o}{\omega x_o}\right)$$

It may be useful to recall the following identity (proved using Euler's identity as well):

$$\arctan(z) = i \frac{1}{2} \frac{\ln(1 - iz)}{\ln(1 + iz)}$$

2. Consider the following Hamiltonian:

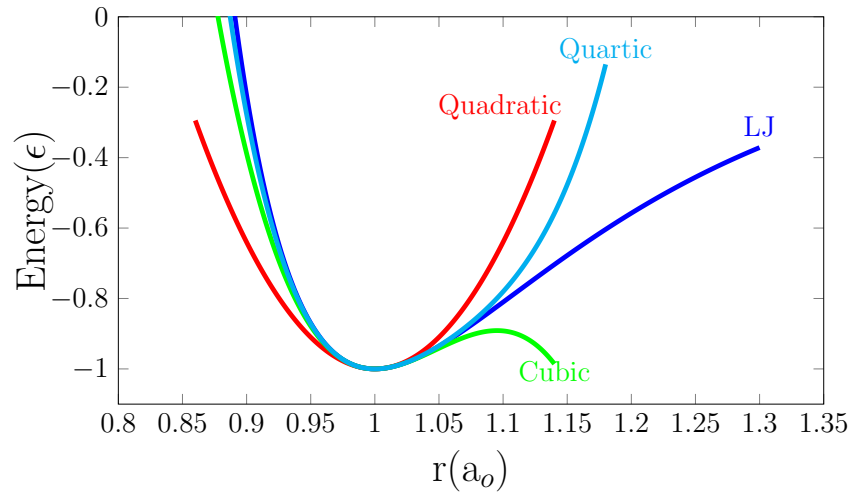
$$H = \frac{p^2}{2m} + \frac{1}{2}\gamma x^2 + \frac{1}{4}\alpha x^4$$

where  $\gamma$  and  $\alpha$  are both positive. Please derive an equation for the following observables to second order in temperature, and sketch each observable as a function of temperature for  $\alpha = 0$  and a small value of  $\alpha$  (a literal sketch is fine):

- The mean square displacement  $\langle x^2 \rangle$ .
  - The heat capacity.
3. Now we will consider a more general Hamiltonian which does have a cubic term: The LJ potential for a dimer (i.e. a pair of atoms, like  $N_2$  or  $O_2$  molecules).

$$v(r) = \epsilon \left[ \left( \frac{a_o}{r} \right)^{12} - 2 \left( \frac{a_o}{r} \right)^6 \right] \\ \approx -\epsilon + \frac{1}{2} v''(a_o)(r - a_o)^2 + \frac{1}{6} v'''(a_o)(r - a_o)^3 + \frac{1}{24} v''''(a_o)(r - a_o)^4$$

- Perform the Taylor series expansion up to 4th order about the minimum of  $r = a_o$ , deriving an equation for  $v'', v''', v''''$  in terms of  $\epsilon$  and  $a_o$ . Below is a picture of the expansion up to second order, up to third order, and up to fourth order.



- (b) Derive an analytic expression for  $\langle x \rangle$ , retaining only the leading order term in temperature. Please make note of the sign of the prefactor (i.e. does the dimer expand or shrink with temperature?).
- (c) Perform molecular dynamics using  $a_o = 1$  and  $\epsilon = 1$ ; and please measure  $\langle x \rangle$  for increasing energies (which is set by the initial displacement, for example). Plot  $\langle x \rangle$  as a function of temperature (or energy), and determine if your measurement is consistent with your perturbation theory results from your previous derivation.